WRIGHT STATE UNIVERSITY

COLLEGE OF ENGINEERING AND COMPUTER SCIENCE

Data Science for Everyone

What is data, and what can we do with it?



By the end of this course you will

1

Know the basics of data science

The 7 steps, types of data, simple statistics and how models predict.

2

Manipulate data yourself

Clean, explore, build features and model data in Orange, with no coding.

3

Use Gen AI ethically

Know its limits, check its answers and protect people's privacy.

How you will be graded

10%

10 quizzes

Short quizzes

25%

Assignment 1

Given Week 3, due Week 6

30%

Assignment 2

Given Week 7, due Week 11

35%

Final project

Weeks 12 to 15

Total: 100%

What is data?

Data is recorded information about the world.

Facts we can store, count and compare.

A number	**A word**	**A picture**	**A sound**	**A place**	**A click**

If it can be recorded, it can be data.

Data comes in many types

Numbers
Today's temperature, your height, a test score

Text
A tweet, a text message, a novel

Images
A selfie, an X-ray, a microscope photo of metal

Audio and video
A voice memo, a song, a security camera clip

Location
Your phone's GPS trail, a delivery route

Behavior
Clicks, likes, steps walked, items bought

Example of data

A table row

Muffin · $2.50 · 9:15 AM · Cash

A text

“The line was long, but the coffee was great!”

A sensor reading

Room temperature: 21.5 °C at 3:00 PM

What is a dataset?

A dataset is data organized as a table.

Student	Major	Hours studied	Coffee cups
Student 1	CS	4	1
Student 2	MechE	2	0
Student 3	Philosophy	5	2
Student 4	Finance	2	3

Row = observation

One student, one sale, one printed part.

Column = variable

One thing we measured about each row.

What is data science?

Using data to answer questions and make better decisions.

Understand the problem → **Find patterns** → **Get results**

Tools and software

Orange | Python | MATLAB

Data science in many domains

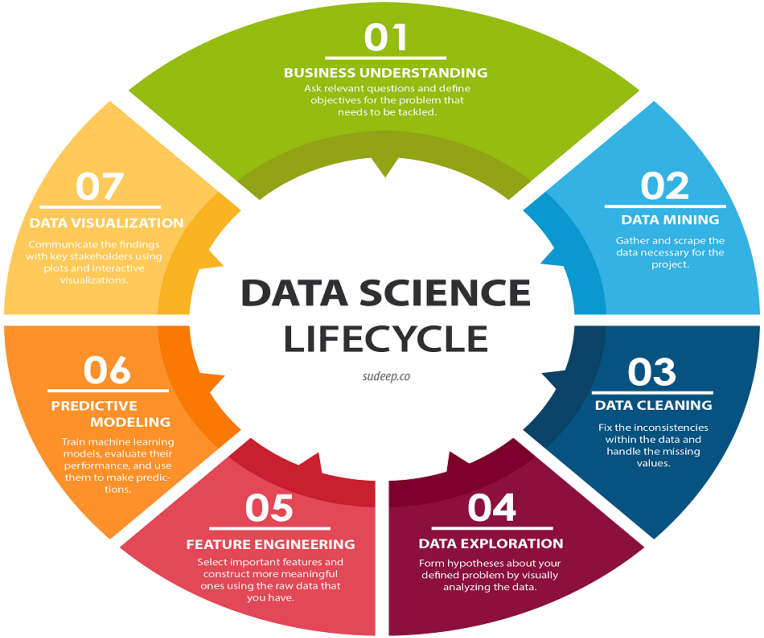
Domain	What data science can do
Manufacturing	Predict defects in 3D printed parts, quality control
Healthcare	Flag patients at higher risk using their record
Digital humanities	Find themes across thousands of books and letters
Sports	Pick lineups and strategy from player statistics
Business	Forecast demand for a product, profit, loss
Environment	Predict floods, heat waves and air quality

You already use data science every day

Where you meet it	What it predicts	Data it learns from
Music and video apps	What you will want next	What you played, skipped and liked
Maps and traffic	How long the trip will take	Speeds and locations from many phones
Email spam filter	Spam or not spam	Words, senders, what people reported
Bank fraud alert	Is this purchase yours?	Your past purchases and locations
Online games	A fair opponent for you	Win rates and skill ratings

A model learns a pattern from past data, then predicts something new.

The data science cycle



1. Business understanding
2. Data mining and extraction
3. Data cleaning
4. Data exploration
5. Feature engineering
6. Predictive modelling
7. Data visualization

One example through the whole cycle

Step	Example: which players will quit a game?
1 Business understanding	Question: find players likely to leave, so we can offer help
2 Data mining and extraction	Get the account table and the play logs
3 Data cleaning	Fix typos, remove duplicate and impossible rows
4 Data exploration	Plot hours played and look for outliers
5 Feature engineering	Build Days Since Last Login, encode Platform
6 Predictive modelling	Predict Yes or No: will this player quit?
7 Data visualization	Chart the quit rate by platform for the manager

Business understanding

Step 1 of 7

What it means

Understand the problem, who it matters to, and what a good answer looks like.

Campus café example

The café throws away muffins every day. Question: how many should we bake each morning? Success: less waste and no sold-out shelves.

We practice this in Weeks 1 and 2 · Made-up example

Data mining and extraction

Step 2 of 7



What it means

Find where the data lives and collect it: public datasets, files, databases, websites, surveys.

Campus café example

Export three months of sales from the cash register. Add the daily weather and the class schedule from public sources.

We practice this in Week 2 · Made-up example

Data cleaning

Step 3 of 7



What it means

Fix wrong, missing and repeated data so we can trust it. Dirty data gives wrong answers.

Campus café example

A register glitch logged −5 muffins sold. One sale appears twice. “Blueberry”, “blueberry ” and “Bluebery” are the same flavor.

We practice this in Week 3 · Made-up example

Data exploration

Step 4 of 7



What it means

Summarize and plot the data to see patterns and surprises before building any model.

Campus café example

Find the average muffins sold per day. A chart shows Fridays are low and rainy days are high.

We practice this in Week 4 · Made-up example

Feature engineering

Step 5 of 7



What it means

A feature is a column a model learns from. Create useful new columns and drop the ones that do not help.

Campus café example

Turn the date into “day of week”. Add “is it raining?”. Drop the receipt number, which tells the model nothing.

We practice this in Weeks 5 and 6 · Made-up example

Predictive modelling

Step 6 of 7



What it means

A model learns patterns from past data and uses them to predict new cases. We always test it on data it has not seen.

Campus café example

Train on two months of sales. Test on the last month. Then predict tomorrow’s muffin sales.

We practice this in Weeks 7 and 9 · Made-up example

Data visualization

Step 7 of 7

What it means

Show the results clearly and honestly so other people can act on them.

Campus café example

One chart of predicted and actual sales by weekday, with a plain recommendation the manager can follow.

We practice this in Week 10 and the final project · Made-up example